RAK19004 WisBlock Green Power Module Datasheet

Overview

Description

RAK19004 is a WisBlock Power module that can harness and convert green power such as wind power, hydroelectric power, or solar power into fixed 5 V output. The module uses a TPS55165-Q1 from Texas Instruments, which is a DC-DC buck-boost converter. Upon startup, the module can have a stable 5 V output from a varying input voltage of 2 V-36 V, and its output current can also be as high as 1 A.

NOTE

- The module has a minimum input voltage of 5.3 V for a startup.
- RAK19004 doesn't have a WisConnector just like the other WisBlock modules, but it is an external module with four (4) mounting holes, so you can place the module in the position that you want.

Features

- 5 V Power Output
- 2 V-36 V Power Input
- Chipset: Texas Instruments TPS55165-Q1
- Module size: 15 X 25 mm

Specifications

Overview

Mounting

The RAK19004 is an external module with four (4) mounting holes, so you can place or mount the module in the position that you want.

Hardware

The hardware specification is categorized into four parts. It discusses the pinouts and their corresponding functions and diagrams of the module. It also covers the electrical and mechanical characteristics that include the tabular data of the functionalities and standard values of the RAK19004 WisBlock Module.

Chipset

Vendor	Part number
Texas Instruments	TPS55165-Q1

Pin Definition

The RAK19004 have two (2) connectors:

- Through-hole connector with a 2.0 mm pitch for the **Power Input**.
- SMD connector with a 1.5 mm pitch for the **5V Output**.

> **WARNING**
>
> **ENSURE** to check correctly the polarity of the cable plugged into the Power Input connector.

Electrical Characteristics

This section shows the maximum and minimum ratings of the RAK19004 module and its recommended operating conditions. Refer to the table presented below.

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VINP, VINL	Supply voltage at VINP and VINL pins (after wake-up)	2	-	36	V
VOUT	Output voltage	-	5	-	V
VOS_FB	Input voltage on VOS_FB pin	0	-	5	V
IGN	Input voltage on IGN pin	0	-	5	V
PG_DLY	Output voltage at PG pin	0	-	5	V
PS	Input voltage on PS	0	-	5	V
SS_EN	Input voltage on SS_EN	0	-	5	V
IOUT	Output Current	0	-	1	A
Efficiency	Power conversion efficiency	-	-	85	%

Mechanical Characteristic

Board Dimensions

The board dimensions of the RAK19004 module are shown in **Figure 1** below.

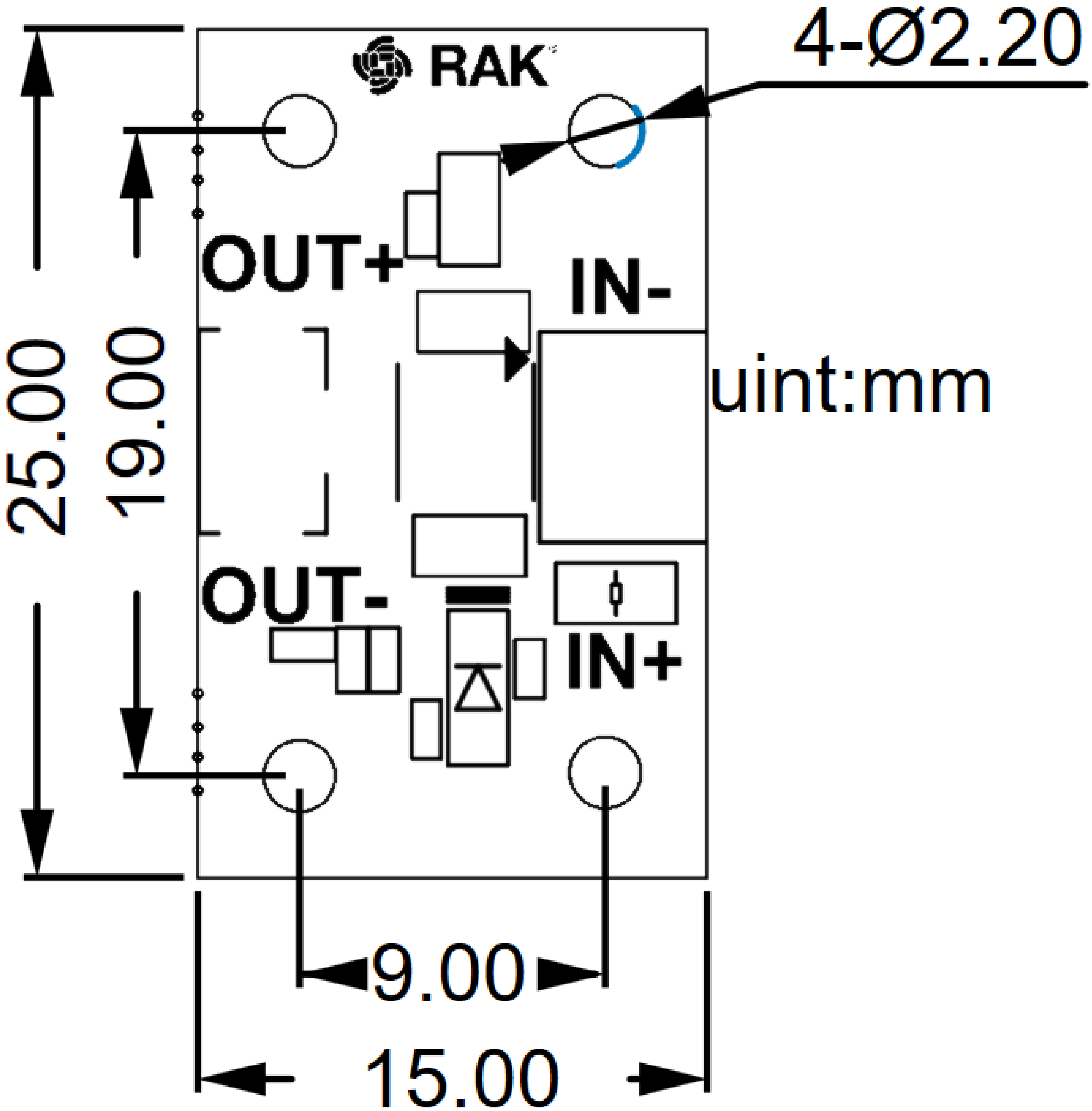

Figure 1: RAK19004 Board Dimensions

> **WARNING**
>
> **ENSURE** to check correctly the polarity of the cable plugged into the Power Input connector.

> **NOTE**

- The connector **J1** close to SILK **POWER_IN** is for Green Power, you can connect it to Green Power.
- The connector **J3** close to SILK **OUT_5V** is for WisBlock Base, you can connect it to the WisBlock Base by using a cable.
- Both **J1** and **J3** are positive poles.

Schematic Diagram

Figure 2 shows the schematic of the RAK19004 module.

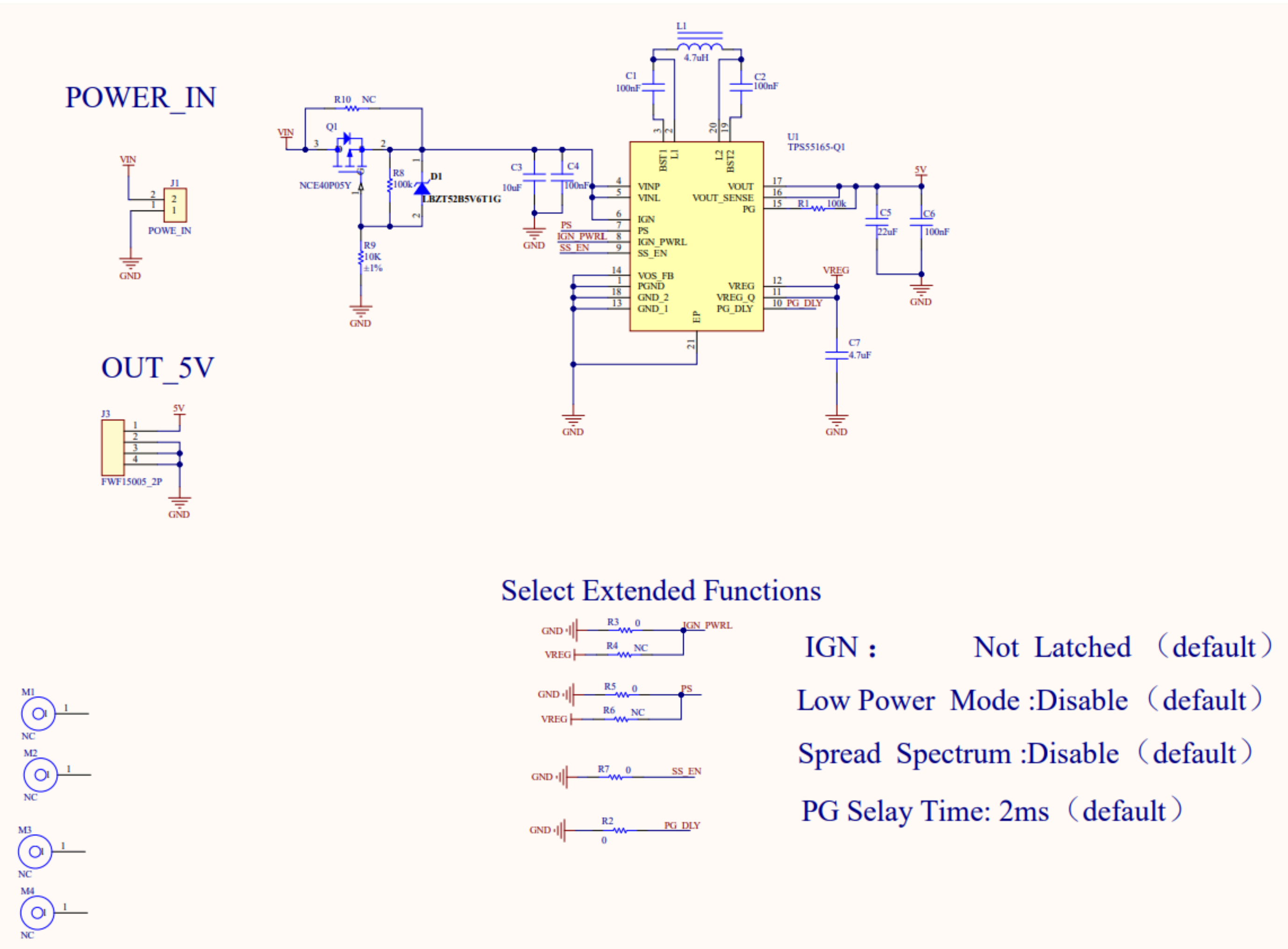

Figure 2: RAK19004 WisBlock Module Schematics

Reverse Connection Protection

Even if the input is reversed, there will be no damage to the device.

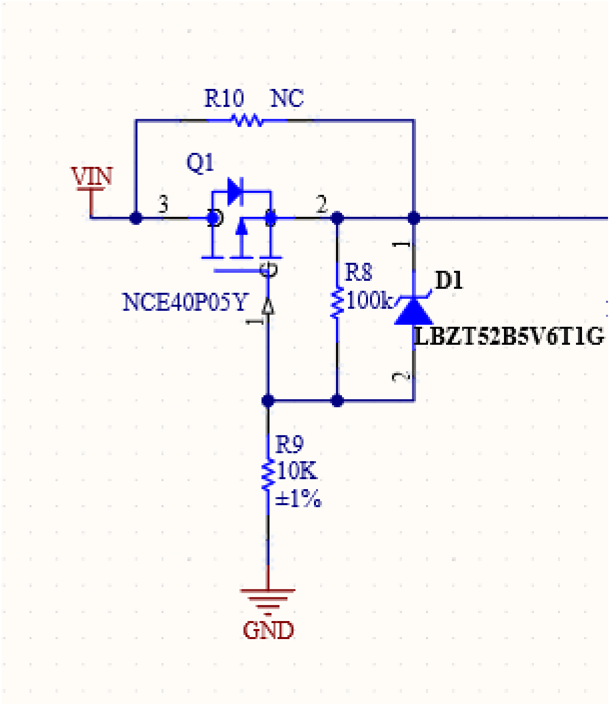

Figure 3: RAK19004 Reverse Protection Diode

Buck-Boost Converter

RAK19004 module uses a TPS55165-Q1 Buck-boost converter.

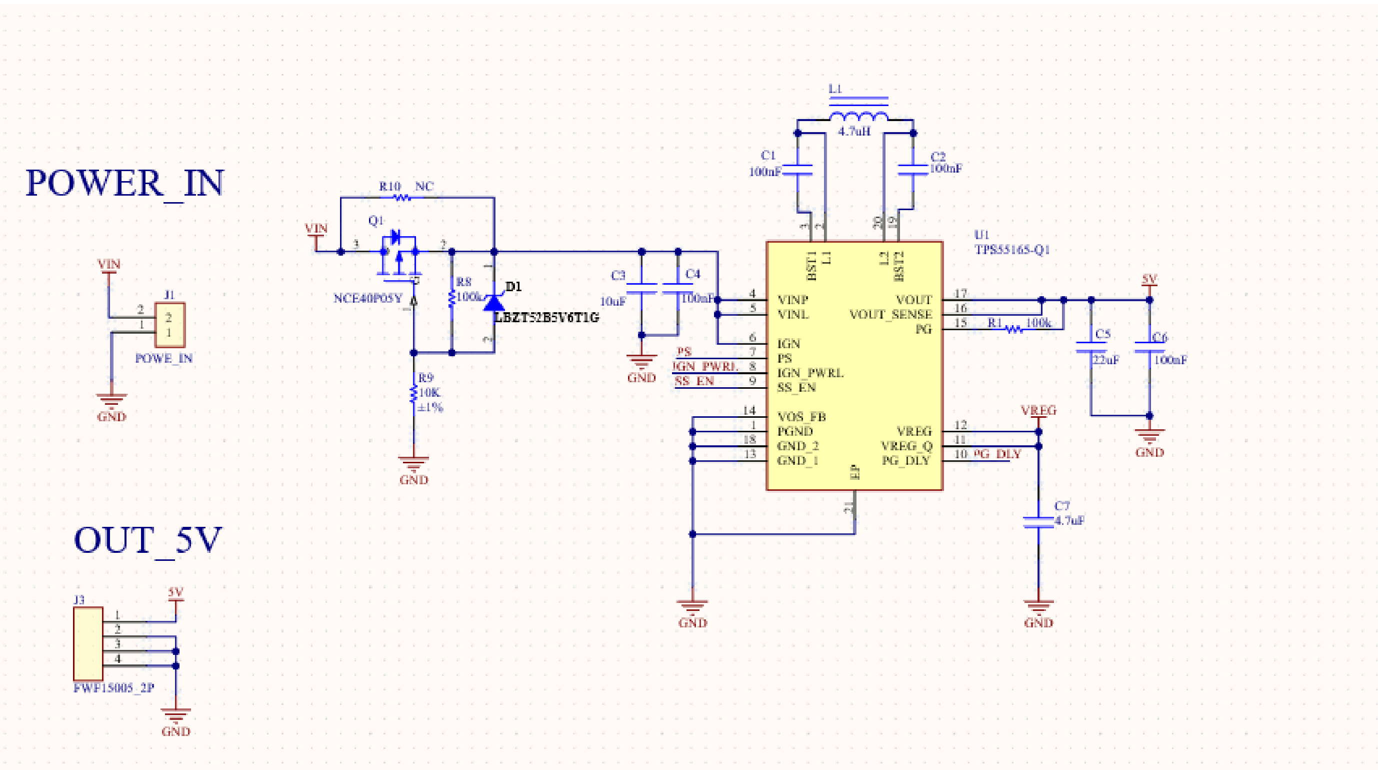

Figure 4: RAK19004 Buck-Boost Schematic

Extended Functions Select

Extended Functions can be selected by placing resistors as an option.

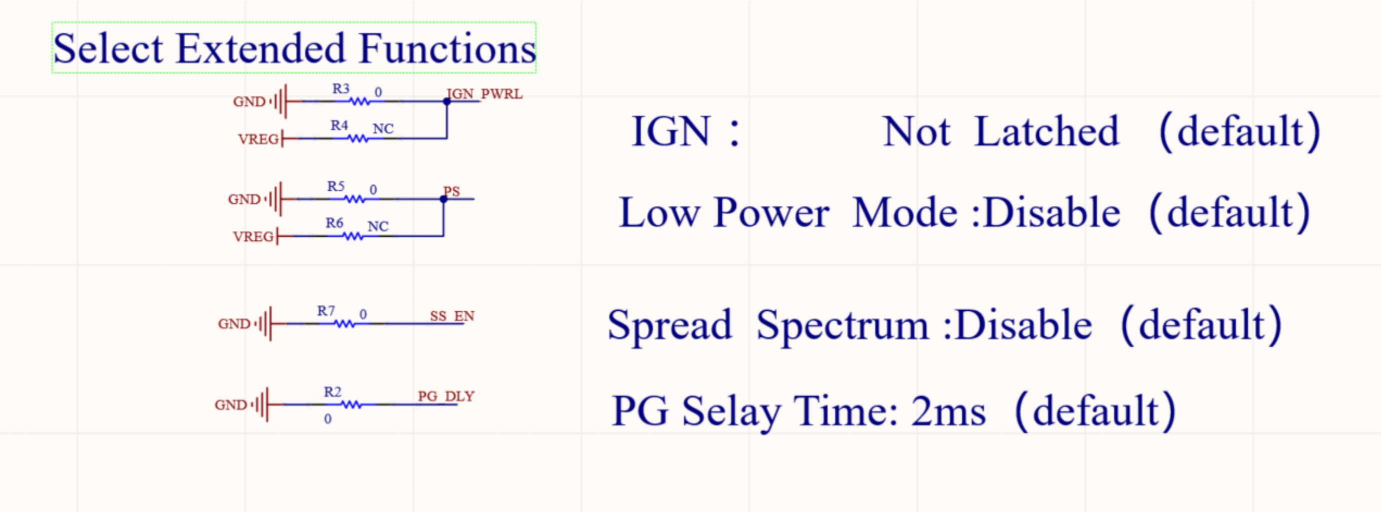

Figure 5: RAK19004 Extended Function Select